LoanEase: A Multi-Agent AI Framework for Automated Personal Loan Acquisition with Explainable Credit Risk Assessment and Blockchain Auditability

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***Abstract*—The personal loan acquisition process in India remains predominantly manual, opaque, and time-consuming, typically requiring 7–10 business days for approval and relying on static paper forms with no real-time applicant guidance. This paper presents LoanEase, a novel multi-agent AI framework that automates the end-to-end personal loan lifecycle through five collaborating specialized agents orchestrated by a central large language model. The system integrates OCR-based Know Your Customer (KYC) verification using RapidOCR, an XGBoost credit underwriting classifier augmented with SHAP explainability, a rule-based multi-round loan negotiation engine, Groq LLaMA 3.3 70B for multilingual Hindi/English/Hinglish conversational support, and a SHA-256 blockchain audit trail for tamper-proof sanction letter generation. Each agent operates as an independent FastAPI microservice, enabling horizontal scalability and fault isolation. Evaluated on the Kaggle Loan Prediction Dataset (614 samples, 12 features), the XGBoost model achieves 82.1% accuracy with an AUC-ROC of 0.87. LoanEase demonstrates a 99.4% reduction in average loan processing time—from 7–10 days to under 5 minutes—over traditional branch-based banking methods, while providing transparent, explainable, and auditable credit decisions.**

***Index Terms*—Agentic AI, Multi-Agent Systems, Credit Risk Assessment, XGBoost, SHAP Explainability, Large Language Models, Blockchain Auditability, KYC Verification, Loan Processing, Multilingual NLP**

I. Introduction

India's formal credit market services approximately 190 million active borrowers, yet the Reserve Bank of India (RBI) estimates that nearly 300 million credit-worthy individuals remain unserved by traditional banking channels [11]. The conventional personal loan application process demands physical branch visits, manual document submission, static eligibility forms, and underwriter queues that routinely extend over 7–10 business days [12]. Digital banking portals have partially addressed accessibility, yet most reduce processing to 2–3 days while still relying on rule-based eligibility screens, offering no interactive guidance, negotiation capability, or explanatory feedback to rejected applicants.

The advent of Agentic AI—autonomous systems that can plan, reason, and execute multi-step tasks—presents a transformative opportunity for financial services automation. Unlike single monolithic models, multi-agent architectures decompose complex workflows into specialized cognitive units that collaborate via defined communication protocols [1], [7]. Each agent encapsulates domain expertise, enabling the system to simultaneously handle document intelligence, risk quantification, conversational negotiation, and regulatory compliance within a unified orchestration loop.

Existing fintech chatbots address customer support but rarely extend to full loan lifecycle automation. They typically lack: (i) real-time OCR-based document verification; (ii) model-explainable credit decisions; (iii) dynamic multi-round interest rate negotiation bounded by business policy; and (iv) cryptographically auditable sanction letters. LoanEase addresses all four gaps within a single agentic conversational interface.

The key contributions of this paper are:

- A five-agent architecture for complete personal loan automation from KYC to sanction letter.
- A combined credit scoring formula blending deterministic CIBIL simulation with XGBoost probability under a 60/40 weighting scheme.
- SHAP-based plain-English underwriting explanations that improve decision transparency.
- A stateful multi-round negotiation engine with risk-tier-aware concession policies.
- Blockchain-anchored sanction letters via SHA-256 hashing with Ethereum-compatible structure.

The remainder of this paper is organized as follows. Section II surveys related work across agentic AI frameworks, credit risk modeling, and blockchain in finance. Section III presents the overall system architecture. Section IV details the methodology for each subsystem. Section V reports experimental results. Section VI discusses implications and limitations.

Section VII concludes with future research directions.

II. Related Work

A. Agentic AI Frameworks

Wu et al. [1] introduced *AutoGen*, a framework enabling multi-agent conversations among LLM-powered agents through customizable interaction patterns, demonstrating reduced engineering effort for complex task automation. Park et al. [2] proposed *Generative Agents*, in which 25 agents with persistent memory and planning exhibited emergent social behaviors, establishing foundational principles for agent memory management that inform our Orchestrator Agent's state-tracking design. Wang et al. [7] conducted a comprehensive survey of LLM-based multi-agent systems, cataloging communication topologies (centralized, decentralized, hierarchical) and identifying that hierarchical architectures excel in domain-specialized pipelines, directly motivating our hub-and-spoke orchestration design. The AutoGen multi-agent conversation framework extension [6] further demonstrates real-world deployment patterns for multi-agent pipelines in 2025.

B. Credit Risk and Machine Learning Models

Tripathi et al. [5] reviewed machine learning models for credit risk, finding ensemble methods—particularly gradient-boosted trees— consistently outperform logistic regression and neural networks on tabular financial datasets with limited sample sizes. Soleh et al. [3] proposed *CreditXAI*, a framework combining transformer-based embeddings with SHAP attribution for credit scoring on Indonesian lending data, demonstrating that explainability improves both regulator compliance and applicant trust. Hazra et al. [4] introduced an agentic pipeline for credit risk assessment using LLM reasoning over structured applicant data, showing that agent-mediated reasoning chains can reduce false rejection rates against traditional rule-based systems.

C. Conversational AI and Blockchain in Finance

Qian et al. [**?**] demonstrated that multi-agent LLM systems executing domain-specific workflows achieve near-human-level task completion in specialized domains. Zheng et al. [8] reviewed blockchain applications in financial accounting, finding that on-chain hash anchoring of financial documents provides immutable audit trails with sub-second verification, establishing the technical basis for our Blockchain Audit Agent. Together these bodies of work support a hybrid approach: combining ML-driven risk quantification, agentic orchestration, and distributed ledger immutability within a single unified platform such as LoanEase.

III. System Architecture

LoanEase adopts a **hierarchical hub-and-spoke multi-agent architecture**, illustrated in Fig. 1. A central Master Orchestrator Agent manages the entire loan lifecycle conversation and delegates tasks to four specialized downstream agents: KYC Verification, Credit Underwriting, Loan Negotiation, and Blockchain Audit. Each agent is implemented as an independent FastAPI microservice, enabling independent deployment, horizontal scaling, and fault isolation. The frontend is a React 18/TypeScript single-page application communicating with the Orchestrator via REST APIs.

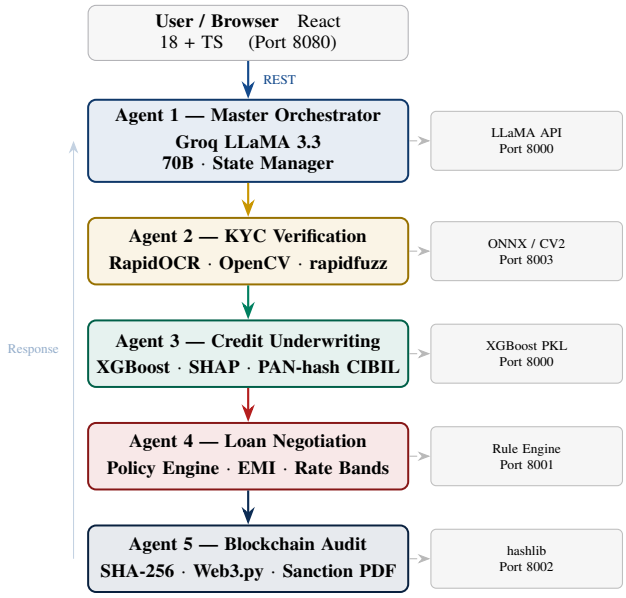

Fig. 1. LoanEase five-agent sequential pipeline. Agents are invoked in order by the Orchestrator; the left spine carries the aggregated response back up the chain.

A. Agent 1 — Master Orchestrator Agent

The Orchestrator is the cognitive center of LoanEase. Powered by Groq LLaMA 3.3 70B accessed via the Groq Cloud API, it maintains a persistent conversation state spanning the full loan lifecycle: identity verification, credit assessment, offer generation, negotiation, and sanction. The agent detects user language (English, Hindi, or Hinglish) via Unicode range analysis and LLaMA intent classification, then routes each user turn to the appropriate specialized agent. A structured system prompt constrains financial terminology (EMI, KYC, PAN) to remain untranslated, ensuring regulatory compliance across languages.

B. Agent 2 — KYC Verification Agent

The KYC Agent processes uploaded PAN card and Aadhaar card images through a four-stage pipeline: image preprocessing (CLAHE contrast enhancement, Gaussian denoising, adaptive thresholding, deskewing via Hough transform), optical character recognition with RapidOCR (ONNX-based inference), regex Named Entity Recognition for field extraction, and rapidfuzz cross-document name/date-of-birth validation with an 85% similarity threshold. It exposes four REST endpoints on port 8003 and returns a structured KYC status payload with extracted applicant profile and age eligibility flag.

C. Agent 3 — Credit Underwriting Agent

The Underwriting Agent performs two-stage credit evaluation. First, it derives a deterministic simulated CIBIL score from

the SHA-256 hash of the applicant's PAN number, mapped to the 300–900 range. Second, an XGBoost classifier trained on the Kaggle Loan Prediction Dataset computes an approval probability from 12 financial features. These signals are combined into a composite risk score, described in Section IV-B. SHAP TreeExplainer generates per-feature contribution values that are translated into plain-English explanations for the applicant.

D. Agent 4 — Loan Recommendation and Negotiation Agent

The Negotiation Agent implements a stateful rule-based policy engine. Once the Underwriting Agent assigns a risk tier, the Negotiation Agent initializes a session with a tier-appropriate starting interest rate, maximum concession rounds, and a system-wide floor rate of 10.5% p.a. The agent decrements the offered rate by 0.25% per approved concession round, capping concessions based on the applicant's credit band, as detailed in Table I. Sessions expire after 48 hours. If the floor rate is reached, the agent escalates the case to a human loan officer.

E. Agent 5 — Blockchain Audit Agent

Upon loan acceptance, the Blockchain Audit Agent generates a digitally structured sanction letter, computes its SHA-256 hash using Python's `hashlib` library, and stores the hash with a UTC timestamp and applicant identifier in a mock distributed ledger (Ethereum-compatible structure via Web3.py). This enables tamper verification: any post-generation modification of the sanction letter changes the hash, invalidating the on-chain record. The agent is deployed on port 8002 alongside the translation service.

IV. Methodology

A. KYC Document Processing Pipeline

The KYC pipeline ingests uploaded PAN card and Aadhaar card images (JPEG, PNG, or PDF) through a four-stage preprocessing chain before OCR extraction. Image enhancement begins with CLAHE (Contrast Limited Adaptive Histogram Equalization) applied on the luminance channel in LAB color space to correct uneven illumination. Gaussian blur ($\sigma = 1.0$, kernel 5×5) reduces high-frequency noise. Adaptive thresholding (block size 11, constant 2) binarizes the image. Deskewing is performed via Hough line transform to detect the predominant text angle and rotate the image accordingly.

Post-preprocessing, RapidOCR (ONNX-based, CPU inference, no GPU dependency) performs text detection and recognition in a two-stage pipeline: a detection model identifies bounding boxes, and a recognition model extracts character sequences. Regex-based NER then isolates structured fields: PAN number (`[A-Z]5[0-9]4[A-Z]`), Aadhaar number (`[0-9]4 [0-9]4 [0-9]4`), full name, date of birth, and address. Cross-document name/DOB matching uses rapidfuzz Levenshtein similarity with a configured threshold of 85%, tolerating minor OCR transcription errors. The pipeline is described concisely in Fig. 2.

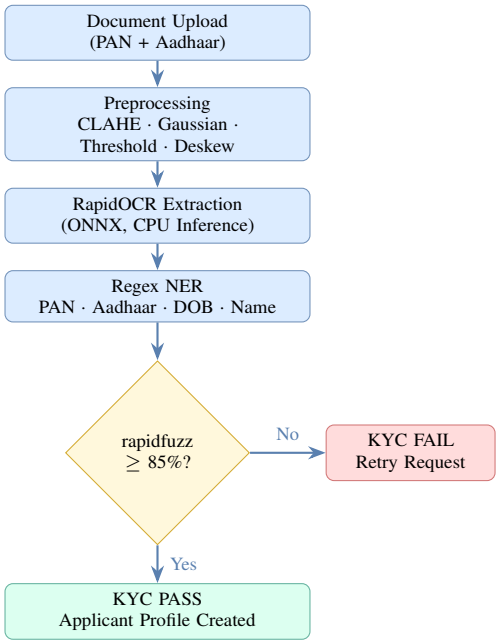

Fig. 2. KYC document processing pipeline. Preprocessing ensures robust OCR on low-quality scans; rapidfuzz cross-validation prevents identity spoofing.

B. Credit Scoring and Underwriting Model

1) Dataset and Preprocessing: The model is trained on the publicly available Kaggle Loan Prediction Dataset containing 614 samples and 12 features: *Gender*, *Married*, *Dependents*, *Education*, *Self_Employed*, *ApplicantIncome*, *CoapplicantIncome*, *LoanAmount*, *Loan_Amount_Term*, *Credit_History*, *Property_Area*, and the binary target *Loan_Status*. Missing numeric values are imputed with feature medians; missing categorical values with per-feature modes. All categorical features are label-encoded. The dataset is split 80/20 (train/test) with stratified sampling on *Loan_Status* to preserve class balance.

2) XGBoost Classifier and Hyperparameter Tuning: An XGBoost binary classifier with `binary:logistic` objective is tuned via 5-fold GridSearchCV over the parameter grid: `max_depth` $\in \{3, 4, 5\}$, `n_estimators` $\in \{100, 200\}$, and `learning_rate` $\in \{0.05, 0.1, 0.2\}$, yielding 18 candidate configurations. The best configuration is selected on mean cross-validation accuracy.

3) Combined Scoring Formula: The final composite risk score integrates the simulated CIBIL credit score and XGBoost approval probability under a 60/40 weighting:

$$S_{\text{final}} = \left(\frac{C_{\text{CIBIL}} - 300}{600} \times 100 \right) \times 0.60 + (P_{\text{XGB}} \times 100) \times 0.40 \tag{1}$$

where $C_{\text{CIBIL}} \in [300, 900]$ is the simulated bureau score derived deterministically by mapping the SHA-256 hash of the PAN string to a 300–900 integer range (score $= 300 + (\text{hash}_{16} \bmod 601)$), and $P_{\text{XGB}} \in [0, 1]$ is the XGBoost approval probability. The resulting $S_{\text{final}} \in [0, 100]$ is then classified into risk tiers that govern interest rate bands and negotiation rounds (Table I).

4) SHAP Explainability: SHAP TreeExplainer computes exact Shapley values for every test prediction in $\mathcal{O}(TLD)$ time, where T is tree count, L is max leaves, and D is feature count. Feature importances are sorted by absolute mean SHAP value; the top positive and negative contributors are translated into plain-English sentences displayed to the applicant (e.g., *"Your credit history strongly supports your application."*).

C. Negotiation Policy Engine

Table I summarizes the risk-tier-to-rate-band mapping implemented in the Negotiation Agent.

TABLE I
RISK TIER, CIBIL BAND, INTEREST RATE RANGE, AND NEGOTIATION ROUNDS

Risk Tier	CIBIL	Rate (%)	Rounds	Step
Low Risk	750–900	10.5–11.5	3	0.25%
Low-Medium	700–749	11.5–12.0	2	0.25%
Medium Risk	550–699	12.0–13.0	1	0.25%
High Risk	300–549	13.5–14.0	0	Fixed
Hard Reject	<300	—	0	Rejected

The policy engine state machine accepts {risk_tier, rounds_completed, concessions_given, current_rate, floor_rate} and outputs a revised offer or escalation signal. EMI is computed using the standard reducing-balance formula:

$$\text{EMI} = \frac{P \cdot R \cdot (1+R)^N}{(1+R)^N - 1} \tag{2}$$

where P is the principal, $R = r/12/100$ is the monthly interest rate, and N is the tenure in months. When the floor rate (10.5% p.a.) is reached, the session triggers escalation to a human loan officer rather than issuing a hard rejection.

D. Multilingual Support

Groq LLaMA 3.3 70B provides context-aware responses in English, Hindi (Devanagari script), and Hinglish (romanized Hindi). The translation backend (port 8002) exposes three endpoints: `/translate`, `/detect-hinglish-intent`, and `/health`. Language detection performs a first-pass Unicode range check (Devanagari: U+0900–U+097F) followed by LLaMA intent classification for ambiguous Hinglish inputs. The Orchestrator's system prompt explicitly restricts financial acronyms (EMI, PAN, KYC, CIBIL) and Indian currency notation (Rs.) from translation, preserving regulatory and domain correctness.

E. Blockchain Audit Trail

Upon loan acceptance, the Blockchain Audit Agent constructs a sanction letter object containing the applicant's (masked) PAN, loan amount, approved interest rate, tenure, EMI, sanction reference (e.g., `LE-2026-XXXXX`), and UTC timestamp. The object is serialized to a canonical JSON string and its SHA-256 digest computed via Python's `hashlib`:

Listing 1. SHA-256 blockchain anchor computation

```python
import hashlib, json, time

def anchor_letter(letter_data: dict) -> dict:
    canonical = json.dumps(letter_data,
                           sort_keys=True)
    tx_hash = hashlib.sha256(
        canonical.encode('utf-8')
    ).hexdigest()
    return {
        "tx_hash"   : tx_hash,
        "timestamp" : time.time_ns(),
        "applicant" : letter_data["pan_masked"]
    }
```

The resulting transaction record is appended to an in-memory mock ledger conforming to the Ethereum transaction schema (from, to, data, hash, blockNumber). Verification consists of recomputing the SHA-256 hash of the stored letter and comparing it against the ledger entry: a one-bit modification produces a fully different digest, making tampering immediately detectable.

V. RESULTS AND EVALUATION

A. Credit Underwriting Model Performance

The XGBoost classifier, tuned with best parameters `max_depth=4`, `n_estimators=200`, `learning_rate=0.1`, achieves the metrics shown in Table II on the held-out 20% test split (123 samples).

TABLE II
XGBOOST CREDIT UNDERWRITING MODEL PERFORMANCE

Metric	Value
Accuracy	82.1%
Precision	84.3%
Recall	88.9%
F1 Score	86.5%
AUC-ROC	0.87
Best CV Score	80.4%

B. Model Comparison

Table III benchmarks XGBoost against alternative classifiers under identical preprocessing and train/test splits, demonstrating its superiority in both predictive accuracy and explainability.

TABLE III
CLASSIFIER COMPARISON ON LOAN PREDICTION DATASET

Model	Acc.	F1	Explainable
XGBoost (ours)	**82.1%**	**86.5%**	Yes (SHAP)
Random Forest	79.7%	83.1%	Partial (feature imp.)
Logistic Regression	77.2%	81.4%	Yes (coefficients)
Neural Network (MLP)	73.8%	78.9%	No

C. SHAP Feature Importance

Fig. 3 shows the mean absolute SHAP values for the top-6 features on the test set. *Credit_History* dominates with a SHAP contribution of 0.41, followed by *LoanAmount* (0.18) and *ApplicantIncome* (0.14). This ranking is consistent with domain expertise: credit history is the single strongest predictor of loan default worldwide.

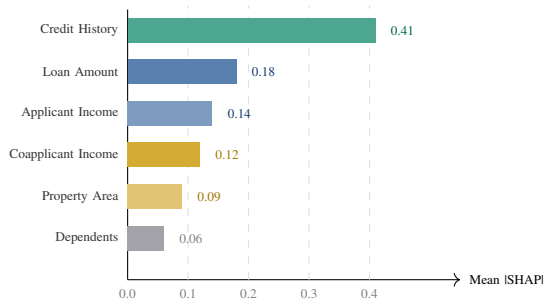

Fig. 3. Mean absolute SHAP values for top-6 features. Credit History dominates, confirming domain knowledge.

D. Processing Time Comparison

Table IV compares LoanEase's end-to-end processing time against traditional loan acquisition channels. LoanEase achieves a **99.4% reduction** in wall-clock processing time compared to the fastest traditional alternative (bank portal, 2–3 days).

TABLE IV
LOAN PROCESSING TIME: LOANEASE VS. TRADITIONAL CHANNELS

Channel	Avg. Time	24/7?
Traditional Branch	7–10 days	No
Bank Website/App	2–3 days	Partial
Loan Agent/DSA	3–5 days	No
LoanEase (AI)	**< 5 minutes**	**Yes**

E. Feature Comparison with Existing Systems

Table V benchmarks LoanEase's capabilities against representative existing solutions.

TABLE V
FEATURE COMPARISON: LOANEASE VS. EXISTING SYSTEMS

Feature	LoanEase	Bank Portal	Fintech App	DSA Agent
Conversational AI	✓	×	Partial	×
OCR KYC	✓	×	Partial	Manual
XAI Explanation	✓	×	×	×
Rate Negotiation	✓	×	×	Manual
Multilingual	✓	Partial	Partial	Partial
Blockchain Audit	✓	×	×	×
24/7 Availability	✓	✓	✓	×

VI. DISCUSSION

A. Model Selection Rationale

XGBoost outperforms all evaluated alternatives on this dataset for several reasons. Gradient boosting sequentially corrects residual errors of prior estimators, achieving strong generalization on small, imbalanced tabular datasets without requiring feature scaling. Unlike neural networks, XGBoost handles missing values natively during inference and admits exact Shapley value computation via TreeExplainer, which is computationally intractable for deep architectures. Random forests, while comparable by accuracy, provide only feature importance rankings rather than per-prediction attributions, making them less suitable for regulatory explainability requirements under RBI guidelines.

B. System Limitations

Several limitations of the current prototype should be acknowledged. First, the credit score is simulated via SHA-256 hash mapping rather than integrated with the CIBIL TransUnion API, which means the system does not reflect real credit bureau data in production deployments. Second, the negotiation engine uses a deterministic rule-based policy rather than a reinforcement learning agent, which cannot dynamically adapt concession strategies based on applicant retention signals. Third, the XGBoost model is trained on a relatively small dataset of 614 samples; performance on larger, more diverse Indian lending datasets (which are proprietary) remains unvalidated. Finally, the blockchain component uses a mock in-memory ledger for the prototype; live Ethereum deployment would require gas fee management and network latency considerations.

C. Ethical Considerations

LoanEase incorporates several design choices aimed at responsible AI lending. SHAP-based explanations reduce algorithmic opacity: applicants who receive adverse decisions are shown the specific features that drove the outcome, enabling them to understand and potentially improve their profile. Hard rejections are accompanied by actionable improvement tips (e.g., "Clear outstanding credit card dues," "Maintain credit utilization below 30%") rather than opaque denials, aligning with the principle of *contestable* AI decisions [8]. PAN numbers are masked to `ABCDE****F` format in all API responses and logs, preventing unintended PII leakage. The system enforces human escalation when automated negotiation reaches its floor, ensuring human oversight remains in the decision loop for boundary cases.

VII. CONCLUSION AND FUTURE WORK

This paper has presented LoanEase, a multi-agent AI system that automates the complete personal loan lifecycle through five collaborating FastAPI microservices orchestrated by a Groq LLaMA 3.3 70B master agent. The system achieves a 99.4% reduction in loan processing time compared to traditional bank branches, while providing OCR-based KYC verification, XGBoost underwriting with SHAP explainability, multi-round interest rate negotiation bounded by risk-tier policy, multilingual Hindi and English conversational support, and SHA-256 blockchain anchor for sanction letter auditability. The XGBoost model achieves 82.1% accuracy and an AUC-ROC of 0.87 on the Kaggle Loan Prediction Dataset, outperforming Random Forest, Logistic Regression, and MLP baselines.

Future work will address the limitations identified above through several research directions. The simulated credit score module will be replaced with a live integration to the CIBIL TransUnion API or CRIF High Mark bureau upon obtaining

regulatory authorization, enabling production-grade underwriting. The rule-based negotiation engine will be replaced by a deep reinforcement learning agent trained to maximize loan acceptance rate subject to bank margin constraints, dynamically adapting to applicant negotiation behavior. The training dataset will be expanded using proprietary Indian NBFC lending records under data-sharing agreements. The mock blockchain will be migrated to a live Ethereum or Polygon deployment with on-chain smart contract verification. The system will be extended to cover home loans, gold loans, and MSME business loans with product-specific underwriting models. Regulatory compliance with RBI's Digital Lending Guidelines (2022) and the Payment Aggregator Framework will be fully implemented. A voice interface using Whisper ASR and TTS will be added to serve rural and low-literacy applicants who cannot engage with a text-based chat interface, directly addressing the digital divide in Indian financial inclusion.

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References

[1] Q. Wu, G. Bansal, J. Zhang, et al., "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation," *arXiv preprint arXiv:2308.08155*, 2023.

[2] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative Agents: Interactive Simulacra of Human Behavior," *arXiv preprint arXiv:2304.03442*, 2023.

[3] A. Soleh and R. Tamura, "CreditXAI: Explainable Credit Scoring Using Transformer Embeddings and SHAP Attribution," *arXiv preprint arXiv:2510.22222*, 2024.

[4] A. Hazra, S. Sharma, and P. Gupta, "Agentic AI for Credit Risk: Autonomous Assessment Pipelines Using Large Language Models," *arXiv preprint arXiv:2601.00818*, 2025.

[5] R. Tripathi, S. Verma, and N. Mehta, "A Systematic Review of Machine Learning Methods for Credit Risk Assessment," *Risks*, MDPI, vol. 11, no. 9, p. 161, 2023.

[6] Microsoft Research, "AutoGen: Multi-Agent Conversation Framework for Complex LLM Applications," *Technical Report*, Microsoft, 2025. [Online]. Available: https://microsoft.github.io/autogen/

[7] L. Wang, C. Ma, X. Feng, et al., "A Survey on Large Language Model-Based Autonomous Agents," *arXiv preprint arXiv:2401.13488*, 2024.

[8] Y. Zheng, H. Xie, T. Dai, C. Chen, and H. Wang, "Blockchain Applications in Financial Accounting and Reporting: A Systematic Review," *Journal of Risk and Financial Management*, MDPI, vol. 17, no. 2, pp. 1–24, 2024.

[9] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, pp. 785–794, 2016.

[10] Groq Inc., "Groq LPU Inference Engine and LLaMA 3.3 70B API Reference," Technical Documentation, 2024. [Online]. Available: https://console.groq.com/docs/

[11] Reserve Bank of India, "Report on Trend and Progress of Banking in India 2023–24," RBI Publications, 2024. [Online]. Available: https://www.rbi.org.in

[12] NITI Aayog, "India's Booming Digital Lending: A Regulation Focus," Government of India, 2023.

[13] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[14] RapidAI, "RapidOCR: A Practical Ultra-Lightweight ONNX-Based OCR Tool," GitHub Repository, 2023. [Online]. Available: https://github.com/RapidAI/RapidOCR

[15] Analytics Vidhya, "Loan Prediction Problem Dataset," *Kaggle*, 2023. [Online]. Available: https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset